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(An Autonomous Institute, with NAAC "A" Grade)

Affiliated to DBATU, RTMNU & MSBTE Mumbai

Department of Artificial Intelligence

"A Place to Learn, A Chance to Grow"

Session: 2025-26



VISION

To be recognized for excellent engineering, developing global leaders both in educational and research in the domain of computer science and wireless engineering.

MISSION

1. To create self-learning environment by facilitating leadership qualities, team spirit and ethical responsibilities.
2. To improve department-industry collaboration, interaction with professional society through technical knowledge and internship program.
3. To promote research and development with current techniques through well qualified resources in the area of computer science and wireless engineering.

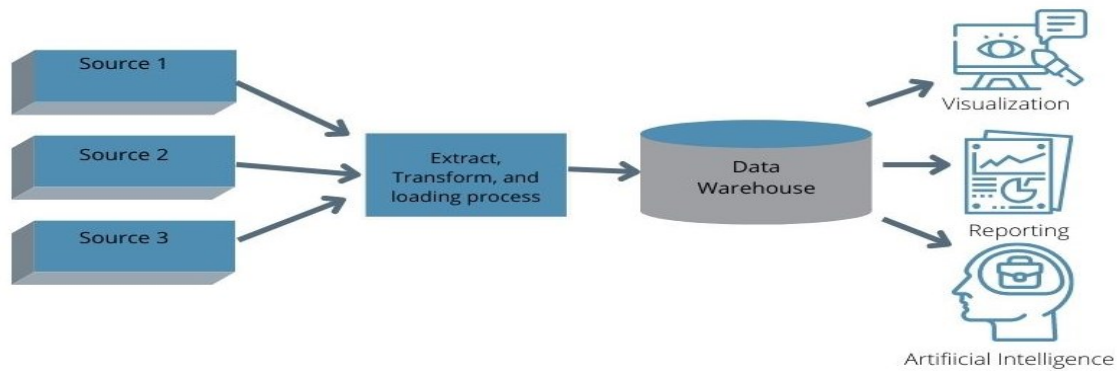
UNIT-01

Data Warehousing and Business Analysis: Data warehousing Components, building a Data warehouse, Data Warehouse Architecture, DBMS Schemas for Decision Support, Data Extraction, Cleanup, and Transformation Tools, Metadata, reporting, Query tools and Applications, Online Analytical Processing (OLAP), OLAP and Multidimensional Data Analysis.

Data Warehousing

- A Data Warehouse is a centralized repository that stores large volumes of structured and sometimes Unstructured data collected from multiple sources.
- It is designed for query and analysis rather than transaction processing.
- A Data Warehouse (DW) is a centralized system that stores large amounts of structured data from various sources, optimized for analysis, reporting, and decision-making.
- Unlike transactional databases, which handle daily operations, a data warehouse focuses on analytical processing.
- Data warehousing is the process of collecting, integrating, storing and managing data from multiple sources in a central repository.
- It enables organizations to organize large volumes of current and historical data for efficient querying, analysis and reporting.
- Data from Different Operational Systems (Like CRM, ERP) is Extracted, Transformed and Loaded (ETL) into the data Warehouse for Business Intelligence (BI) Purpose.



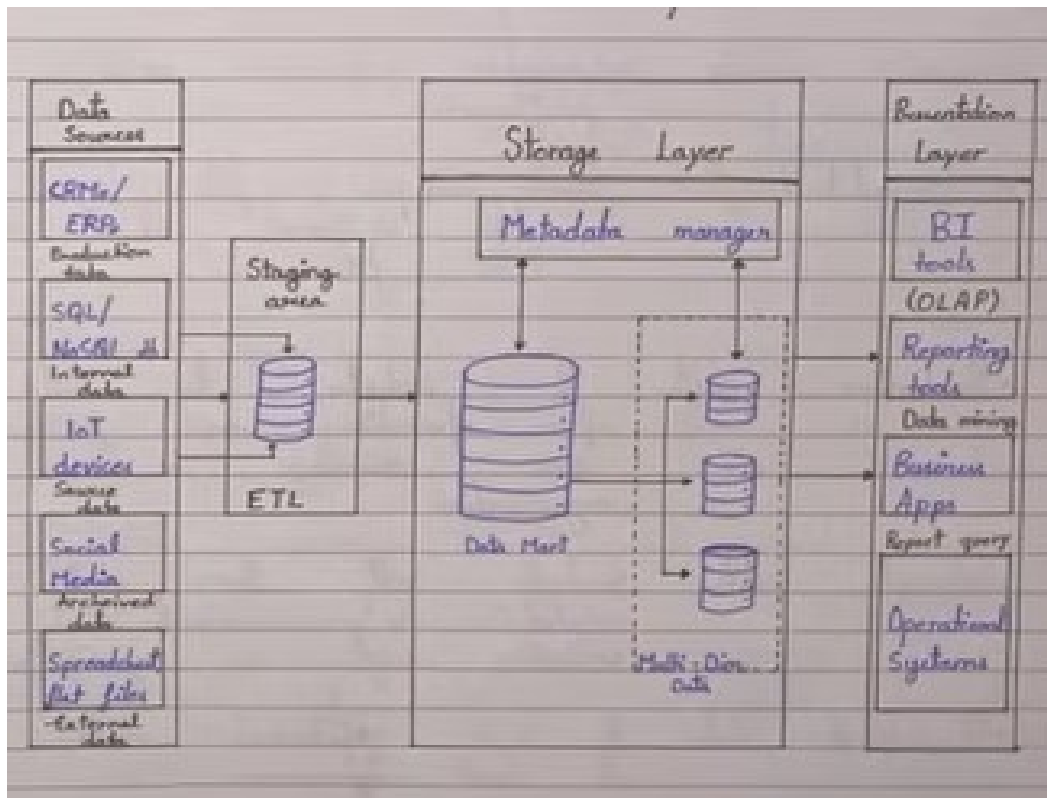
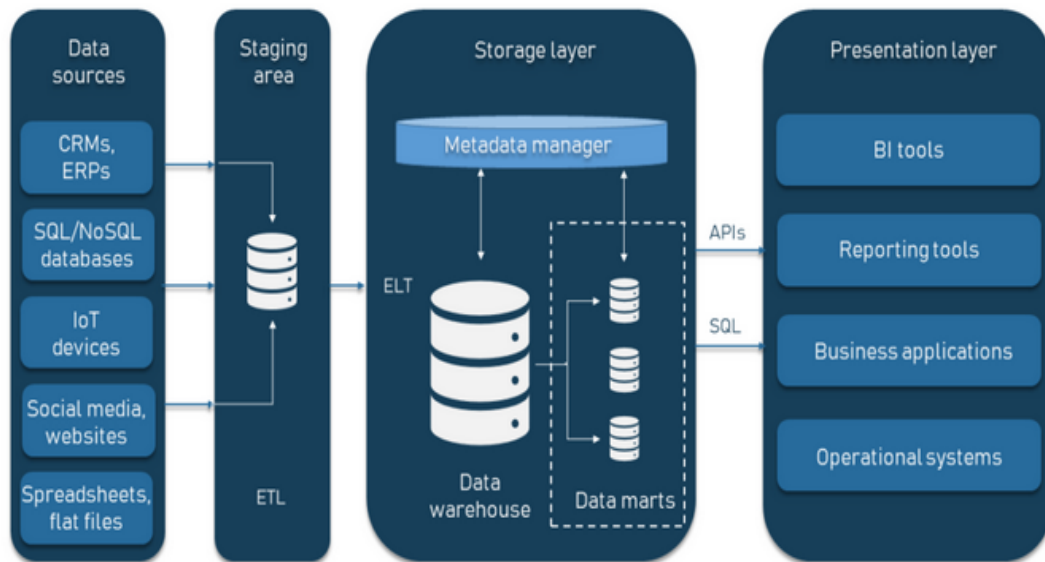


Data Warehouse vs DBMS

Database	Data Warehouse
An organized collection of related data which stores data in a tabular format.	A central location which stores consolidated data from multiple databases.
It Contains detailed data.	It Contains summarized data.
Generally, a Database stores current and up-to-date data which is used for daily operations.	A Data Warehouse maintains historical data over time. Historical data is the data kept over years and can used for trend analysis, make future predictions and decision support.
It Uses Online Transactional Processing (OLTP).	It Uses Online Analytical Processing (OLAP).
A database is generally application specific.	A Data Warehouse is integrated generally at the organization level, by combining data from different databases.
Helps to perform fundamental operations of a business.	Helps to analyze the business.
It processes short term data	It processes long term data
Constructing a Database is not so expensive.	Constructing a Data Warehouse can be expensive.
Example: A database stores related data, such as the student details in a school.	Example: A data warehouse integrates the data from one or more databases , so that analysis can be done to get results , such as the best performing school in a city.

Components of Data Warehouse

A data warehouse is an integrated system that consolidates data from operational systems and external sources, providing valuable insights for decision-making. It includes various components that work together to store, manage and access data efficiently. The data moves from the data source area through the staging area to the presentation server. The entire process is better known as ETL (extract, transform and load).



A) Data Sources: These are the various operational systems, databases and external data feeds that provide raw data to be stored in the warehouse. Serve as the primary input for the data warehouse.

- a) **CRM / ERP Systems:** Customer Relationship Management, Enterprise Resource Planning System.
- b) **SQL / NoSQL Databases:** Structured and Unstructured Databases.
- c) **IoT Devices:** Data collected from IoT sensors and devices.
- d) **Social Media:** Information from social platforms.
- e) **Spreadsheets / Flat Files:** Simple file Formats like excel sheets or CSV Files.

Types of Data:

- i. **Production Data:** It Includes the Data of Operations and Production's Running in an Organization.
- ii. **Internal Data:** Includes all the related to internal Activities of an Organizations.
- iii. **Archived Data:** It is historical data; all past information is stored in achieved data.
- iv. **External Data:** It is not inside the Company's Internal data but it has all external data like marketing Strategies etc.

B) Data Staging:

- Data Staging is an **ETL process** where, Data is **Extracted** from multiple sources, Data is **Transformed** (cleaning, formatting) and Data is **Loaded** into the data warehouse.
- The ETL process is responsible for extracting data from different sources, transforming it into a suitable format and loading it into the data warehouse.

C) Storage Layer:

- It Co-ordinates all the Services and activities of data warehouse.
 - a) **Metadata Manager:** It Handles Information about data. Metadata describes the structure, source and usage of data within the warehouse, making it easier for users and systems to understand and work with the data.
 - b) **Data Marts:** It is a Smaller Focused upset of a data warehouse. It provides relevant data to specific users, making it faster and easier to access data for their needs.
 - c) **Multi-dimensional Data:** It is structured to represent data in multiple dimensions, often visualized as a data cube. It allows users to analyze data from various perspectives, making it easier to perform OLAP (Online Analytical Processing), which helps in decision making.

D) Presentation Layer:

- a) **OLAP (Online Analytical Processing) Tools:** OLAP tools allow users to analyze data in multiple dimensions, providing deeper insights and supporting complex analytical queries. Method used to analyze multi-dimensional data from various perspectives, enabling complex queries and reporting.
- b) **Data Mining:** Process of discovering patterns, trends, and useful insights from large datasets using statistical and ML techniques.
- c) **Report Query:** A specific step or command to retrieve and display information from a database.

Information from a Database

- **BI Tools (Business Intelligence Tools):** Software applications that analyze data and provide insights for decision making.
- **Reporting Tools:** Applications that generate structured reports from data.
- **Business-Related Operational Systems:** Systems that users use to handle data for business operations.

Building a Data Warehouse

Building a data warehouse is the process of collecting, integrating, storing, and managing data from multiple heterogeneous sources to support decision-making, reporting, and analysis

There are **10 major steps** to build a data warehouse.

- 1. Gather Business Requirements:** This step focuses on understanding business needs, It involves defining key data metrics and identifying the data that needs to be analyzed for decision making.
- 2. Identify and Analyze Data Sources:** In this step, various data sources such as operational databases, files, and external systems are identified that will feed data into the data warehouse.
- 3. Design the Data Warehouse Schema:** Choose a schema model to define how data will be organized. A suitable schema model (Star schema, Snowflake schema, etc.) is selected to define how data will be organized in the warehouse.
- 4. Choose the Right ETL Tools:** ETL tools are selected to efficiently extract data from sources, transform it into the required format, and load it into the data warehouse. warehouse.
- 5. Create ETL Process:** The ETL pipeline is created and automated to extract, clean, transform, and load data into the data warehouse.
- 6. Build Data Storage Layer:** Databases are set up to store the transformed data in an organized manner according to the designed schema.
- 7. Implement Metadata Management:** Organize metadata to manage the structure, data lineage, and how the data is used, ensuring easy understanding.
- 8. Create Data Marts (Optional):** Build smaller, focused data marts for specific departments or business units.
- 9. Integrate OLAP and BI Tools:** Set up OLAP cubes for multidimensional analysis. Integrate BI tools like Tableau or Power BI for reporting and visualization.

10. Test, Optimize, and Secure: Test for performance, optimize data retrieval processes, and establish security measures to protect the data.

A Three Tier Data Warehouse Architecture

Tier-1: Bottom Tier

Bottom Tier is the foundation of the data warehouse, responsible for collecting, processing and storing data from multiple sources. It plays a critical role in preparing data for analysis.

Key Components:

- **Data Sources:** Includes operational databases (OLTP systems), flat files, spreadsheets, external APIs, CRM/ERP systems and web logs. These are the raw inputs feeding into the data warehouse.
- **Data Storage:** Processed data is stored in a Relational Database Management System (RDBMS) or a multidimensional database designed to support structured querying and analysis.

ETL Process (Extract, Transform, Load)

This is the core function of the bottom tier:

1. **Extract:** Gathers raw data from different, often incompatible sources.
2. **Transform:** Converts data into a consistent format, applying business rules, cleansing errors, handling missing values and resolving duplicates.
3. **Load:** Loads the transformed data into the warehouse, organizing it for fast access and analysis.

Tier-2: Middle Tier

The Middle Tier hosts the OLAP server, which processes complex analytical queries. It acts as a bridge between the data storage layer (bottom tier) and the user interface (top tier), ensuring data is quickly retrieved, aggregated and ready for reporting and analysis. It is designed for high-speed analytical processing. OLAP server models come in three different categories, including:

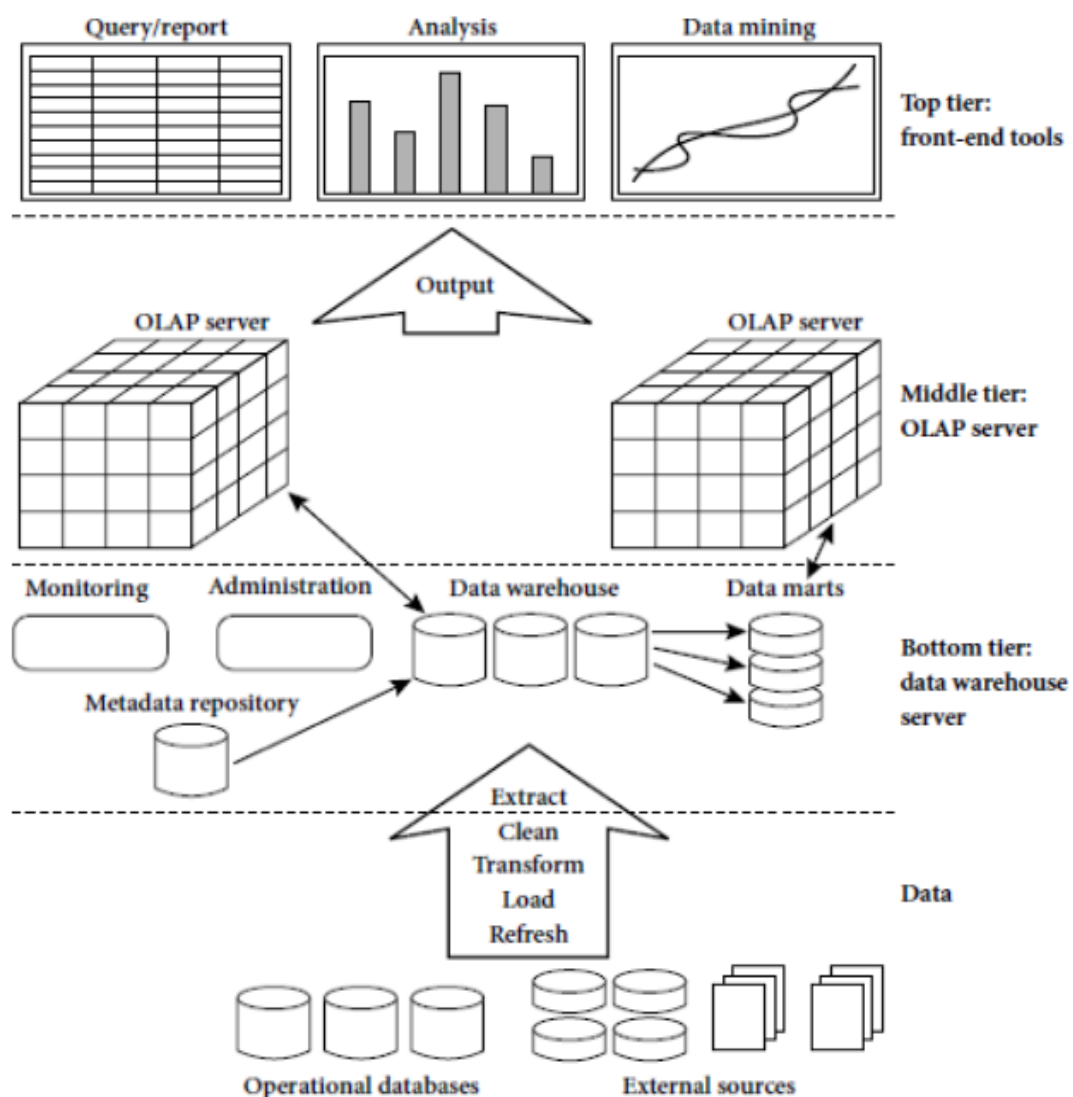
- **ROLAP (Relational OLAP):** This model uses a relational database to store and manage warehouse data. It is ideal for handling large data volumes as it operates directly on relational databases.
- **MOLAP (Multidimensional OLAP):** This model stores data in a multidimensional cube. The storage and retrieval processes are highly efficient, making MOLAP suitable for complex analytical queries that require aggregation.
- **HOLAP (Hybrid OLAP):** It is combination of relational and multidimensional online analytical processing paradigms. HOLAP is the ideal option for a seamless functional flow across the database systems when the repository houses both the relational database management system and the multidimensional database management system.

Tier-3: Top Tier

The Top Tier in the Three-Tier Data Warehouse Architecture comprises the front-end client layer, which is essential for interacting with the data stored and processed in the lower tiers. This layer includes a variety of business intelligence (BI) tools and techniques designed to facilitate easy access and manipulation of data for reporting, analysis, and decision-making.

BI tools are critical components of the Top Tier, providing robust platforms through which users can query, report, and analyze data. Popular BI tools include:

- **IBM Cognos:** Offers comprehensive reporting capabilities.
- **Microsoft BI Platform:** Integrates well with existing Microsoft products, providing a familiar interface for users.
- **SAP BW:** Specializes in managing large datasets and integrating with other SAP products.
- **Crystal Reports:** Known for its powerful reporting features.
- **SAS Business Intelligence:** Provides advanced analytics.
- **Pentaho:** A versatile tool for data integration and visualization.

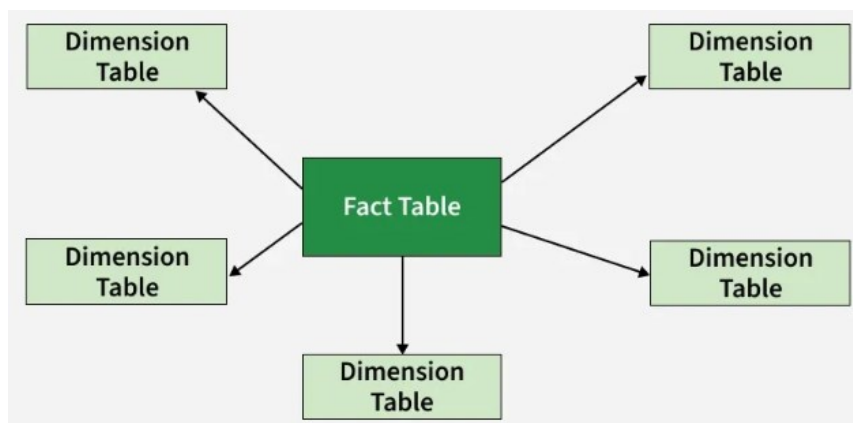


DBMS Schemas for Decision Support

In a Data Warehouse, DBMS schemas define how data is organized and stored to support decision-making and analytical queries. These schemas are specially designed to optimize query performance, data analysis, and reporting. The most commonly used schemas for decision support are Star Schema, Snowflake Schema, and Fact Constellation Schema.

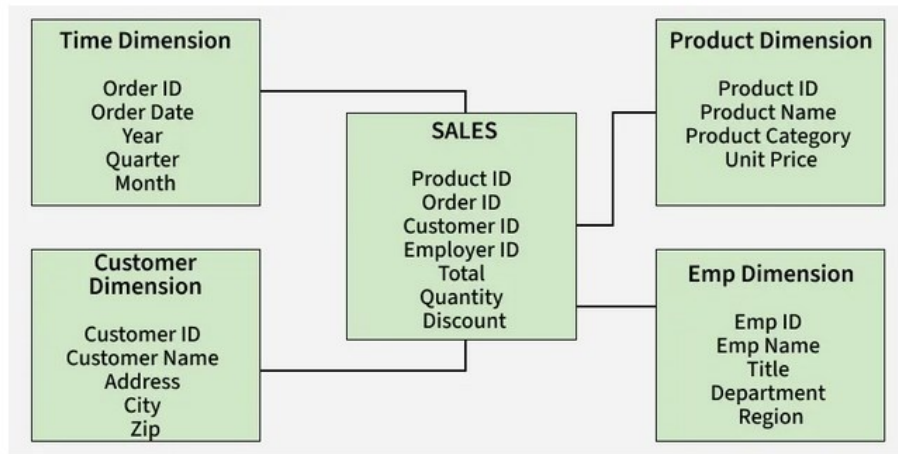
1. Star Schema:

- The star schema is the simplest data warehouse scheme.
- A Star Schema organizes data into a central fact table linked to multiple dimension tables, forming a layout that visually resembles a star.
- In star schema each of the dimensions is represented in a single table. It should not have any hierarchies between dimensions.
- It contains a fact table surrounded by dimension tables. If the dimensions are de-normalized, we say it is a star schema design.
- In star schema only one join establishes the relationship between the fact table and any one of the dimension tables.
- A star schema optimizes the performance by keeping queries simple and providing fast response time. All the information about each level is stored in one row.



Example (Sales Data Warehouse)

To demonstrate how a star schema works, consider a Sales Data Warehouse where each sales transaction is stored in a central fact table and is analyzed through surrounding dimension tables. This model supports fast, flexible analysis across products, customers, time, and employees.

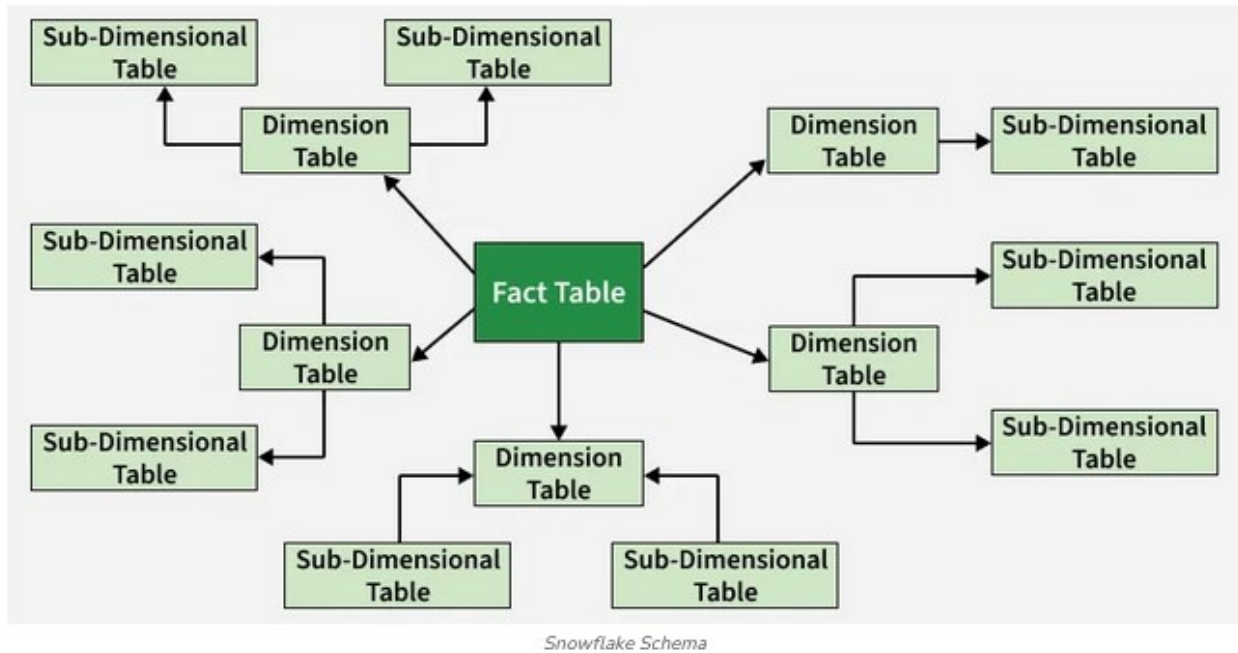


Star Schema Diagram for Sales Data Warehouse

- SALES fact table stores numerical measures such as total amount, quantity, and discount, along with foreign keys to each dimension.
- Surrounding dimension tables provide descriptive attributes that allow the facts to be examined from different perspectives.
- Product Dimension describes what was sold and the Customer Dimension identifies who purchased it.
- Time Dimension indicates when the transaction occurred, and the Employee Dimension specifies who processed the sale.

2. Snowflake Schema:

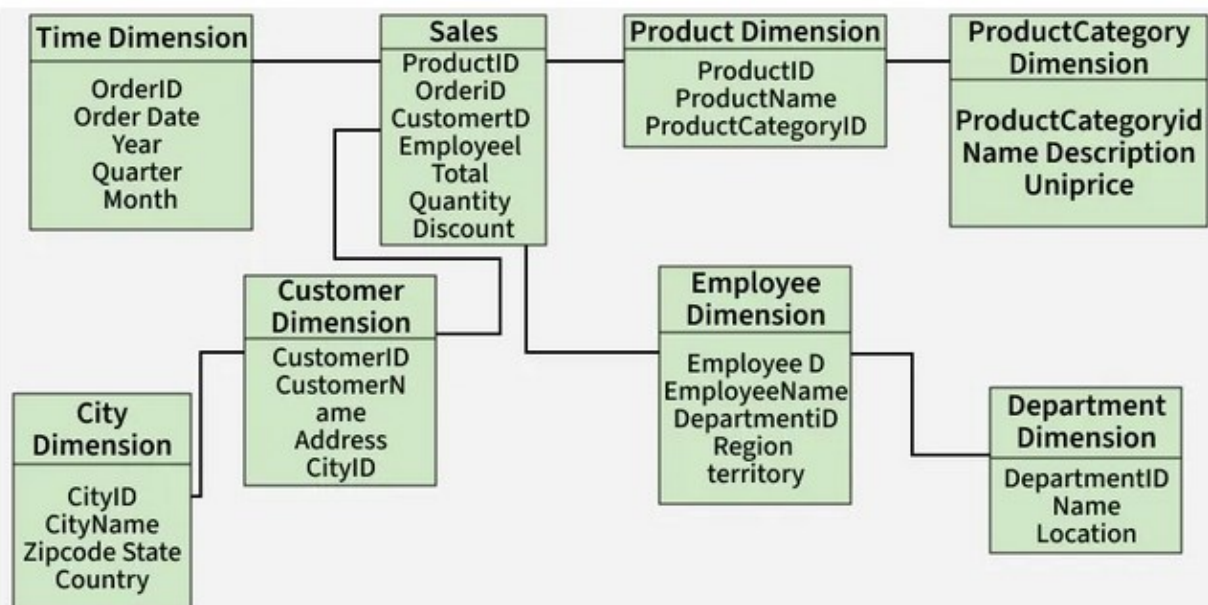
- Snowflake schema is a more complex data warehouse model than a star schema.
- In snowflake schema contains the fact table, dimension tables and one or more than tables for each dimension table.
- Dimension tables are normalized into multiple related tables, creating a hierarchical or "snowflake" structure.
- Fact table is still located at the center of the schema, surrounded by the dimension tables. However, each dimension table is further broken down into multiple related tables, creating a hierarchical structure that resembles a snowflake.
- Snowflake schemas normalize dimensions to eliminate redundancy. The result is more complex queries and reduced query performance.



Example of Snowflake Schema

Below is a simplified representation of how a snowflake schema model looks

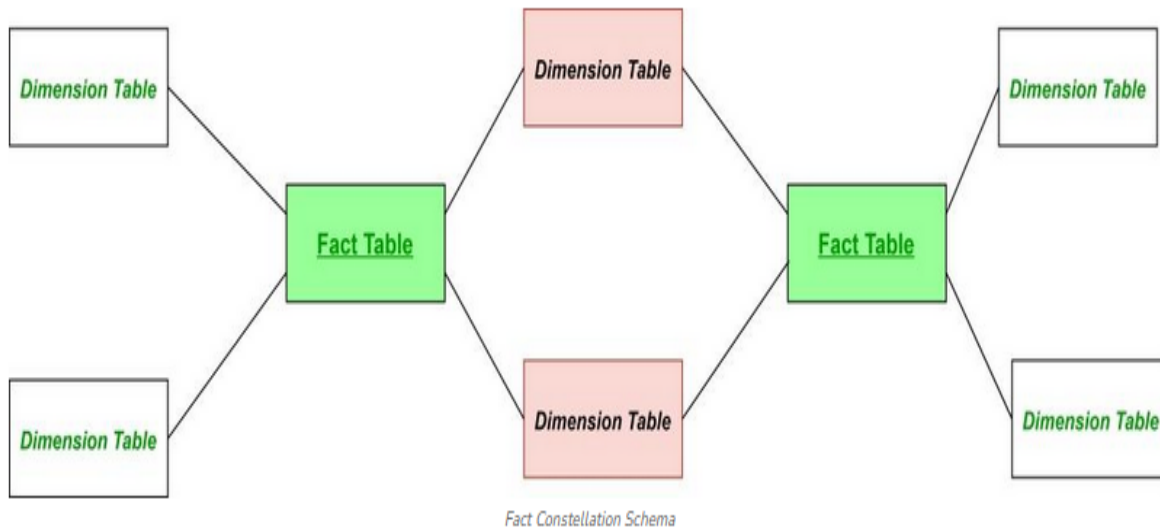
- Employee dimension includes attributes like EmployeeID, Name, DepartmentID, Region, and Territory. DepartmentID links to the Department table, which holds department details like Name and Location.
- Customer dimension includes CustomerID, Name, Address, and CityID. CityID links to the City table, which stores City Name, Zipcode, State, and Country.
- Product dimension includes ProductID, ProductName, ProductCategoryID. ProductCategoryID links to the Product Category Dimension: Category Name, Description, Unit Price.



Example: Snowflake Schema

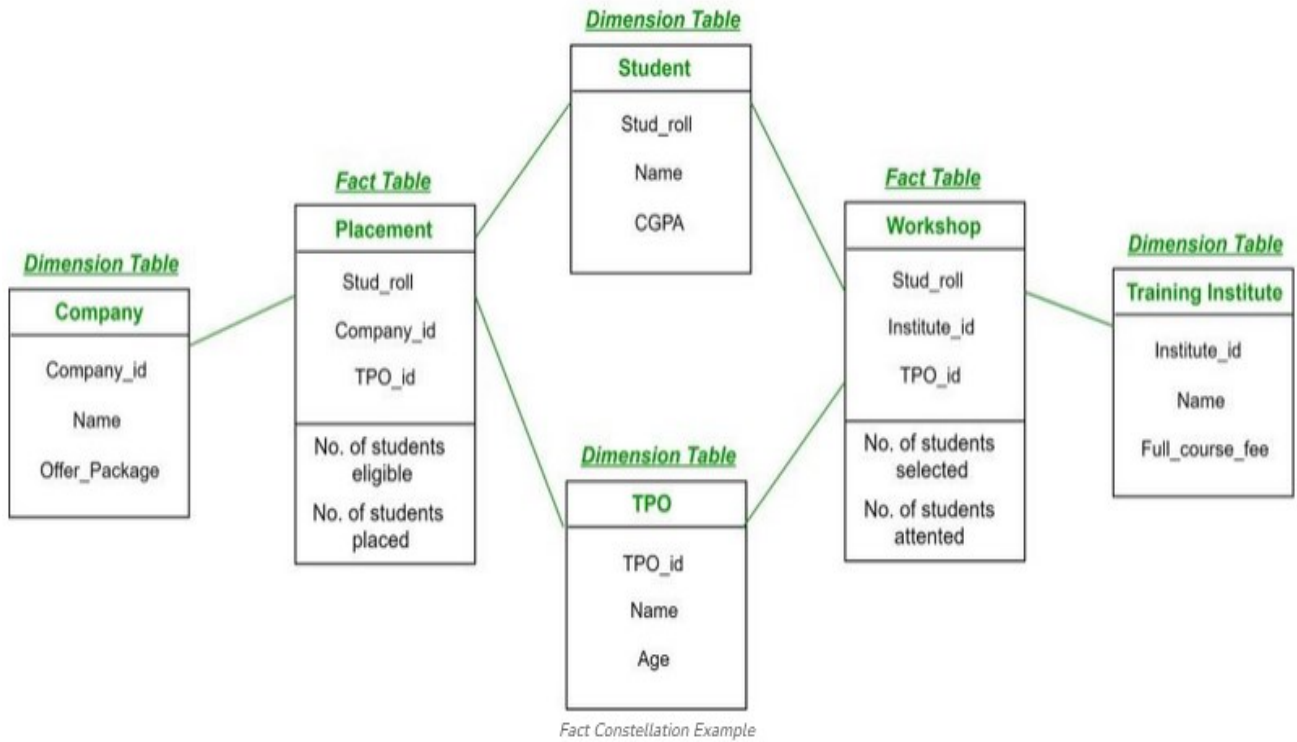
3. Fact Constellation Schema:

- Fact Constellation in Data Warehouse modelling is a schema design that integrates multiple fact tables sharing common dimensions, often referred to as a "Galaxy schema." This approach allows businesses to conduct multi-dimensional analysis across complex datasets.
- Unlike simpler models like the Star Schema and Snowflake Schema, the Fact Constellation Schema consists of multiple fact tables that share common dimensional tables.
- Unlike the snowflake schema, the constellation schema, in fact, uses heavily complex queries to access data from the database.
- Fact constellation schema consists of multiple fact tables sharing common dimension tables. It is used for complex decision support systems.



Example

In the diagram below, we have two fact tables: Placement and Workshop, each associated with their own star schemas. These star schemas share common dimension tables, forming the Fact Constellation schema, also referred to as the Galaxy Schema.



- The Placement fact table records data related to student placements. It connects to the **Company**, **Student**, and **TPO** dimension tables.
- The Workshop fact table captures information related to workshops attended by students. It is connected to the Student, TPO, and Training Institute dimension tables.
- The Placement fact table connects to the Company, Student, and TPO dimension tables, forming a star schema. The Workshop fact table connects to the Student, TPO, and Training Institute dimension tables, forming another star schema. Both star schemas share the Student and TPO dimension tables, thus combining into a Fact Constellation Schema.
- This structure allows for complex analysis across different aspects of the student data, such as placement and workshop participation, using shared dimensions. The fact tables (Placement and Workshop) represent different business processes, and by sharing common dimensions (like Student and TPO), this schema offers a powerful way to analyze and process data in a data warehouse.

The difference between Snowflake Schema and Fact Constellation Schema:

S.NO	Snowflake Schema	Fact Constellation
1.	Snowflake schema contains the large central fact table, dimension tables and sub dimension tables.	While in fact constellation schema, dimension tables are shared by many fact tables.
2.	Snowflake schema saves significant storage.	While fact constellation schema does not save storage.
3.	The snowflake schema consists of one star schema at a time.	Whereas the fact constellation schema consists of more than one star schema at a time.

S.NO	Snowflake Schema	Fact Constellation
4.	In snowflake schema, tables can be maintained easily.	In fact constellation schema, the tables are tough to maintain.
5.	Snowflake schema is a normalized form of star schema.	While fact constellation schema is a normalized form of snowflake schema and star schema.
6.	Snowflake schema is easy to operate as compared to fact constellation schema as it has less number of joins between the tables.	Fact constellation schema is not easy to operate as compared to snowflake schema as it has multiple number of joins between the tables.
7.	In snowflake schema, to access the data from database simple and less complex query is used.	While in fact constellation schema, to access the data from database heavier complex query is used.

Data Extraction, Cleanup, and Transformation Tools

Data Extraction:

- Data extraction is the critical first step in the ETL (Extract, Transform, Load) process, which moves data from various source systems into a data warehouse.
- Extraction is the service of extracting information from a source system for additional help in a data warehouse environment.
- Its primary goal is to retrieve the necessary data with minimal impact on the source systems' performance.
- Data extraction can be done manually, using tools like SQL or Excel, or it can be automated using specialized software or scripts.

Types of Data Extraction

Data can be extracted from many sources, not limited to digital data, like text on hard copy documents can be read using **Optical Character Recognition (OCR)** scanners and converted to digital text. But among digital types of data extraction, the following are the common methods:

- **Structured data extraction:** This type of extraction is used to extract data that is stored in a structured format, such as a database or spreadsheet. This type of data is often easy to extract because it is organized in a tabular format, with well-defined columns and rows.
- **Semi-structured data extraction:** This type of extraction is used to extract data that is stored in a semi-structured format, such as XML or JSON. This type of data is more complex than structured data because it does not have a fixed schema, but it is still organized in a way that allows for automated extraction.
- **Unstructured data extraction:** This type of extraction is used to extract data that is stored in an unstructured format, such as a text document or image. This type of data is the most difficult to extract because it does not have a fixed format and may require the use of natural language processing or machine learning techniques.

- **Web scraping:** This is a specific type of data extraction that is used to extract data from websites. This process involves sending an HTTP request to a website, parsing the HTML or XML response, and extracting the desired data.
- **API data extraction:** Some companies and organizations expose their data through Application Programming Interfaces (APIs), allowing for easy data extraction by making requests to the provided endpoint and getting the data in a structured format, like JSON or XML.
- **Data Extraction from Cloud:** With the rise of cloud computing, more companies are storing their data on cloud-based platforms, like AWS, Azure and GCP. These platforms have their own set of APIs and connectors that allows for easy data extraction

Cleanup:

In a data warehouse, Data Cleanup (also known as data cleansing or scrubbing) is the process of detecting and correcting or removing corrupt, inaccurate, or irrelevant records from a dataset. Since data warehouses aggregate information from multiple, often messy sources (CRMs, ERPs, Flat Files), cleanup is essential to prevent "Garbage in, Garbage Out."

The Core Cleanup Tasks

Cleanup typically happens in the **Staging Area** of the ETL/ELT process. Key tasks include:

- **Deduplication:** Identifying and merging multiple records for the same entity (e.g., a customer listed twice because they used two different email addresses).
- **Standardization:** Unifying formats so they are consistent. For example, converting "USA," "U.S.A.," and "United States" all to a single code like "US."
- **Handling Missing Values:** Deciding whether to leave a field null, delete the row, or fill it with a default value (like "Unknown") or a calculated average (imputation).
- **Correcting Structural Errors:** Fixing typos, mislabeled categories, or inconsistent naming conventions (e.g., "M" vs "Male").
- **Filtering Outliers:** Identifying data points that are technically "valid" but clearly wrong (e.g., a birth year of 1850) and removing or flagging them for review.

Transformation Tools

The Transform phase is where the magic happens. Data extracted in the previous phase is often raw and inconsistent. During transformation, the data is cleaned, aggregated and formatted according to business rules. This is a crucial step because it ensures that the data meets the quality standards required for accurate analysis.

Common transformations include:

- **Data Filtering:** Removing irrelevant or incorrect data.
- **Data Sorting:** Organizing data into a required order for easier analysis.
- **Data Aggregating:** Summarizing data to provide meaningful insights (e.g., averaging sales data).

Data warehouse transformation tools automate cleaning, reshaping, and enriching raw data into usable formats for analysis, with popular examples including dbt (SQL-focused), Matillion, Talend, Informatica, and cloud-native solutions like Azure Data Factory and AWS Glue, fitting into ETL/ELT processes to ensure data quality, consistency, and faster insights for BI and machine learning.

Types of Tools

- **ETL/ELT Platforms:** Comprehensive tools handling extraction, transformation, and loading (e.g., Talend, Informatica, Matillion, Hevo).
- **Modern Transformation Tools:** Focus on in-warehouse transformations, often using SQL (e.g., dbt, SQLMesh).
- **Cloud-Native Services:** Integrated with cloud data warehouses (e.g., Azure Data Factory, AWS Glue).
- **Orchestration Tools:** Manage data pipelines (e.g., Apache Airflow) often used with custom Python/SQL scripts for complex transformations.

Online Analytical Processing (OLAP)

Online Analytical Processing(OLAP) refers to a set of software tools used for data analysis in order to make business decisions. OLAP provides a platform for gaining insights from databases retrieved from multiple database systems at the same time. It is based on a multidimensional data model, which enables users to extract and view data from various perspectives.

Type of OLAP servers:

The three major types of OLAP servers are as follows:

- ROLAP
- MOLAP
- HOLAP

1. Relational OLAP (ROLAP):

- Relational On-Line Analytical Processing (ROLAP) is primarily used for data stored in a relational database, where both the base data and dimension tables are stored as relational tables.
- ROLAP servers are used to bridge the gap between the relational back-end server and the client's front-end tools.
- ROLAP servers store and manage warehouse data using RDBMS, and OLAP middleware fills in the gaps.

Advantages:

- It can Handle Huge Amount of Data.
- It Provides Greater Scalability.
- It Efficiently handles Complex Analytical Queries Which Involves Calculations and joins across multiple tables.

Disadvantages:

- Only expertise person can handle ROLAP.

2. Multidimensional OLAP (MOLAP):

- MOLAP (Multidimensional OLAP) uses array-based storage to provide fast, multidimensional analysis.
- Data is stored as a multidimensional array, where each cell corresponds to a combination of dimension values and holds a measured fact. Unlike ROLAP, MOLAP defines all possible dimension combinations, which leads to sparsity-many empty cells consuming space.
- To manage this, MOLAP systems use indexing and hashing for efficient data retrieval.
- All calculations are pre-computed when the cube is built, enabling quick responses to complex queries.
- MOLAP is ideal for slicing, dicing, and high-speed analytical operations.

Advantages:

- MOLAP is easy to use.
- In MOLAP information Retrieval is Fast.
- It is Suitable for Slicing & Dicing Operations.

Disadvantages:

- It can handle Limited amount of Data.
- It cannot deal with detailed data.

3. Hybrid OLAP (HOLAP):

- ROLAP and MOLAP are combined in Hybrid On-Line Analytical Processing (HOLAP). HOLAP offers greater scalability than ROLAP and faster computation than MOLAP. HOLAP is a hybrid of ROLAP and MOLAP. HOLAP servers are capable of storing large amounts of detailed data.
- On the one hand, HOLAP benefits from ROLAP's greater scalability.
- HOLAP, on the other hand, makes use of cube technology for faster performance and summary-type information.
- Because detailed data is stored in a relational database, cubes are smaller than MOLAP.

Advantages:

- HOLAP Provides advantages of both ROLAP & MOLAP.

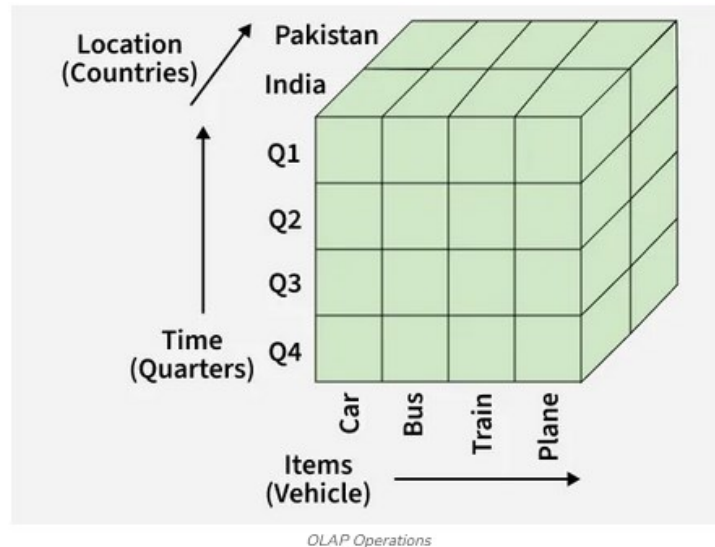
Disadvantages:

- Because of the Support of ROLAP & MOLAP, HOLAP Architecture is Complex.

OLAP Operations:

- OLAP (Online Analytical Processing) is a software technology that enables users to analyze data from multiple database systems simultaneously.
- It is based on a multidimensional data model, where data is represented in the form of cubes, also known as hyper-cubes. Each cube consists of dimensions (e.g., Location, Time, Product) and measures (e.g., Sales, Profit).

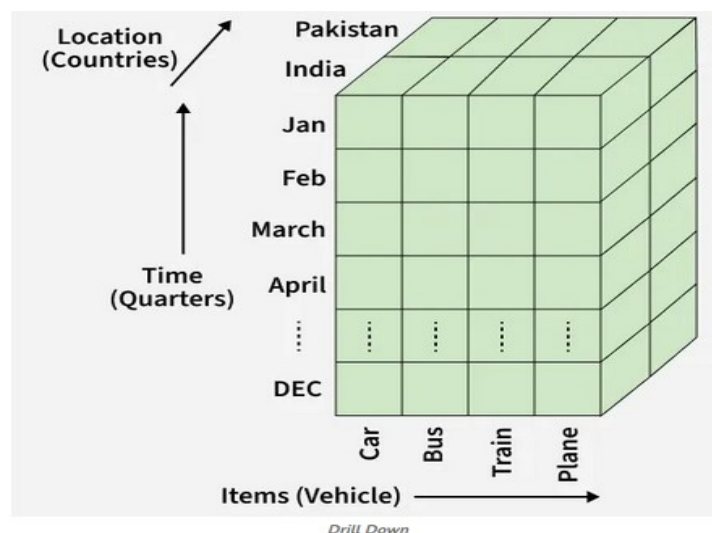
- OLAP is widely used in data warehousing and business intelligence systems to support analytical queries, trend analysis, and decision-making.



OLAP supports five fundamental analytical operations that allow users to view data from different perspectives and levels of detail:

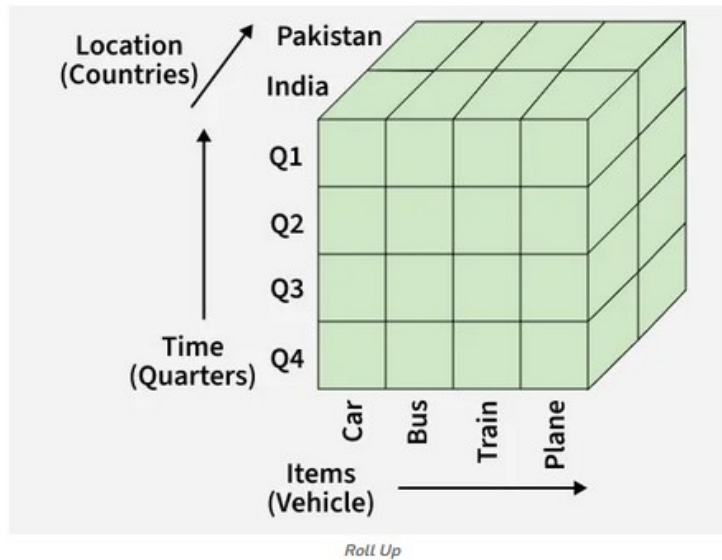
1. Drill Down

- The Drill Down operation provides a more detailed view of the data.
- It moves from a summary level to a lower level in the concept hierarchy (for example, from Year -> Quarter -> Month).
- **How it works:** Moving down in the hierarchy & adding new dimensions for more granularity.
- **Example:** Viewing sales data for 2024 -> Q1 -> January instead of only yearly totals.



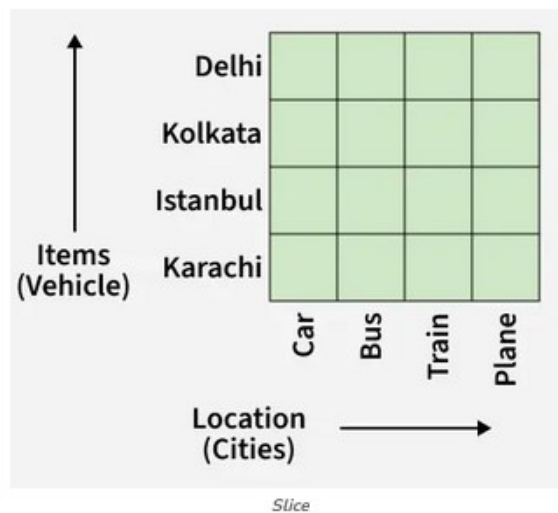
2. Roll Up

- The Roll Up operation is the opposite of Drill Down.
- It aggregates or summarizes data to provide a higher-level overview.
- **How it works:** Climbing up in the concept hierarchy & reducing the number of dimensions.
- **Example:** Aggregating sales data from City -> Country or from Month -> Quarter.



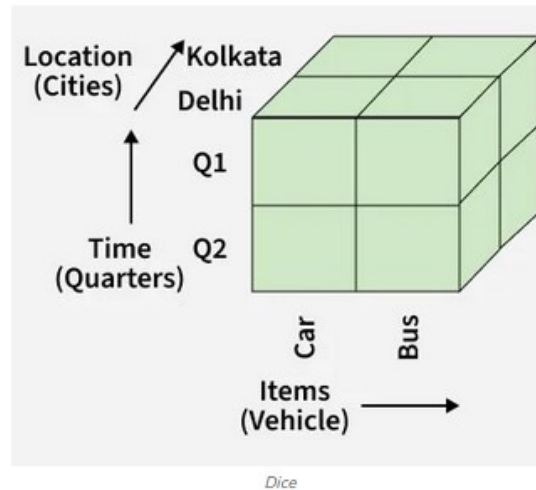
3. Slice

- The Slice operation selects a single dimension from the cube, creating a new sub-cube with reduced dimensionality.
- It helps focus on a specific data slice for analysis.
- **Example:** Selecting Time = "Q1" to analyze sales across all products and regions for the first quarter.



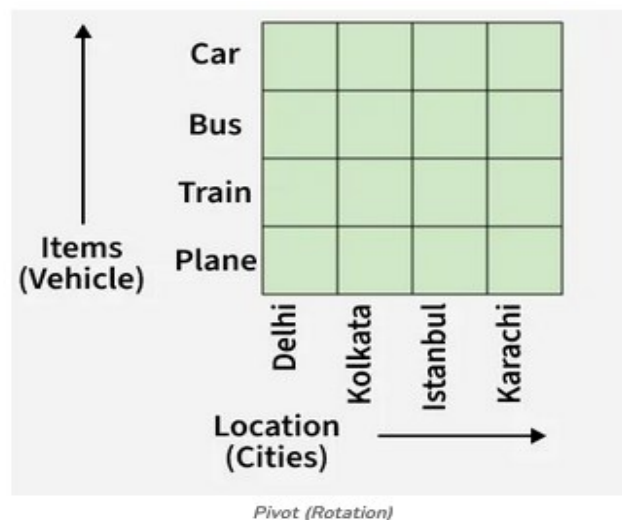
4. Dice

- The Dice operation selects data from the cube by applying filters on two or more dimensions to form a sub-cube.
- This results in a smaller cube focused on these specific dimensions.
- **Example:** Selecting: Location = "Delhi" or "Kolkata", Time = "Q1" or "Q2" & Item = "Car" or "Bus"



5. Pivot (Rotation)

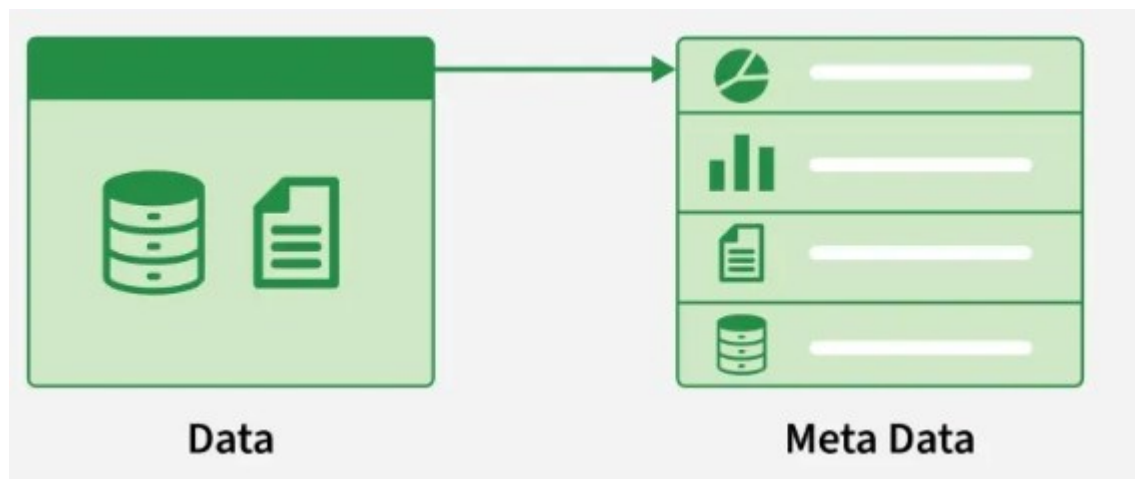
- The Pivot operation (also known as Rotation) reorients the cube to provide a different view of the data.
- It helps users visualize data from different perspectives by rotating rows and columns.
- **Example:** Swapping the Time and Location axes to compare sales by quarter across different regions.



Metadata

- Metadata in a data warehouse is "data about data," acting as a roadmap.
- Metadata in data warehousing is data that describes other data.
- It is data that Provide Information about other data, but not the Content of the Data.
- It provides information about the structure, meaning, origin, and usage of data stored in a data warehouse.
- Instead of containing actual business data, metadata acts as a guidebook that helps users and systems understand how data is organized and how it should be used.
- It helps to Organize, Find and Understand data,

For example: if a customer's purchase amount is the data, then the information about when the purchase was made, the data type, and the source system is its metadata.



Types of Metadata in Data Warehousing

Metadata in a data warehouse is commonly divided into three main categories:

1. Descriptive Metadata:

- It is a type of Metadata that describes the content of a data or digital asset, such as its title, author, date created, and keywords.
- Descriptive metadata helps users discover and identify Different types of assets Through a given set of Keywords.
- Descriptive metadata can include administrative metadata or Structural metadata, as well as other type of data that Someone can use to find or identify Single or multiple Digital assets.

2. Structural Metadata:

- It is a type of Metadata that describes the organization of a data or digital asset, such as the page layout of a document or the chapter structure of an ebook. It is often used to help users navigate and understand the content of an asset.
- This Data gives the Structural Information of the assets. It Indicate how an asset in Organized, Such as page No., Sections, Chapters, Indexes, Table of Contents.

3. Administrative Metadata:

- It is a type of Metadata that describes the management and preservation of a digital asset or data, such as its file format, copyright status, and access restrictions.
- It is often used to ensure the long-term preservation and accessibility of an asset.

4. Technical Metadata:

- It is a type of Metadata that describes the technical characteristics of a data or digital asset, such as its resolution, color space, and file format.
- It is often used to ensure the proper display and playback of an asset.

5. Rights metadata:

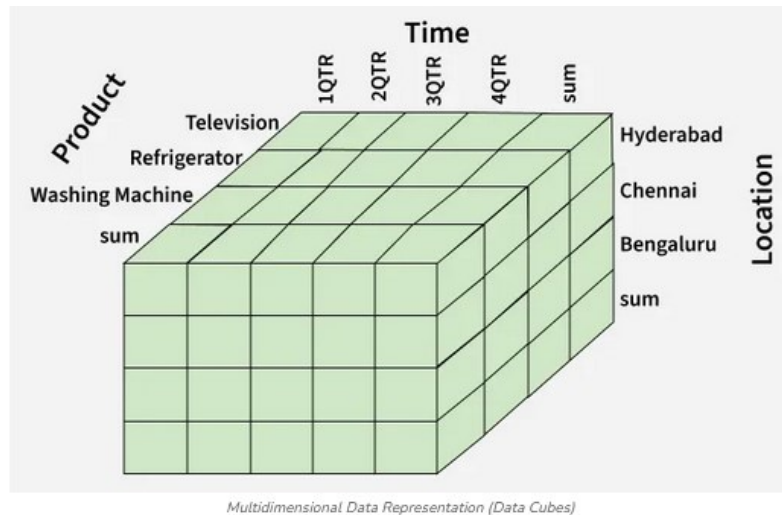
- It is a type of Metadata that describes the rights associated with a digital asset, such as its copyright status, licensing information, and permissions for reuse.
- It is often used to ensure compliance with copyright laws and to facilitate the sharing and reuse of assets.

Multidimensional Data Model:

- A multidimensional data model organizes data into "cubes" with facts (numerical measures like sales) and dimensions (descriptive attributes like time, product, location) to support fast, complex analysis in data warehouses.
- In technical terms, the multidimensional data model in data warehouse structures data into dimensions and facts, making it easy to analyze from different angles quickly.
- It uses fact tables (measures like sales) and dimension tables (context like dates, locations) structured in schemas (Star, Snowflake) to provide fast, aggregated insights for decision-making, unlike traditional row/column database.
- A Multidimensional Data Model (MDM) organizes data into multiple dimensions such as time, product, location to support fast analytical queries in data warehouses and OLAP systems.

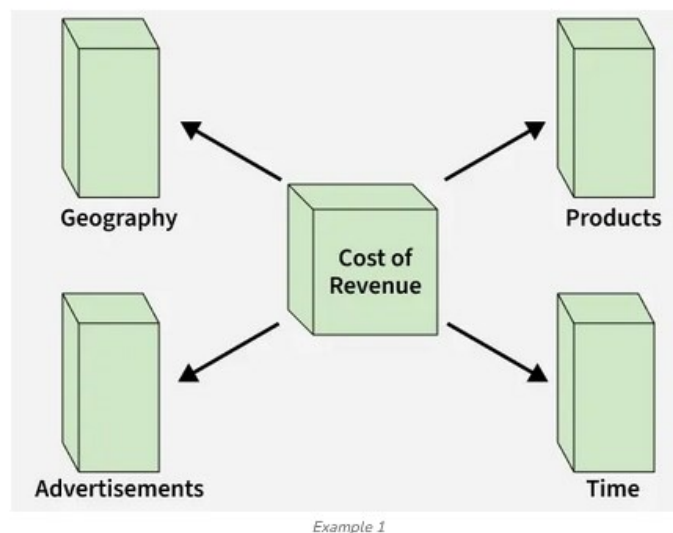
Key Features or Terminologies of the Multidimensional Data Model

- **Supports Multi-Dimensional Analysis:** Enables users to analyze data across several dimensions (e.g., sales by product, region, and time) to identify business patterns and trends.
- **Used in OLAP and Data Warehousing:** Forms the foundation of OLAP systems, where data is stored in multidimensional structures for rapid aggregation and reporting.
- **Data Cubes:** Represents data as cubes that allow viewing information from different perspectives and performing complex analytical operations efficiently.
- **Facts:** Numerical measures (e.g., sales, revenue). Stored in a fact table.
- **Dimensions:** Descriptive attributes (e.g., product, region, customer). Stored in dimension tables. The fact table links to multiple dimensions to support multi-axis analysis.



Example: Understanding the Multidimensional Model

1. Let us take the example of a firm. The revenue cost of a firm can be recognized on the basis of different factors such as geographical location of firm's workplace, products of the firm, advertisements done, time utilized to flourish a product, etc.



2. Let us take the example of the data of a factory which sells products per quarter in Bangalore. The data is represented in the table given below:

Location = "Bangalore"				
Time (quarter)	Type of item			
	Jam	Bread	Sugar	Milk
Q1	350	389	35	50
Q2	260	528	50	90
Q3	483	256	20	60
Q4	436	396	15	40

2D factory data

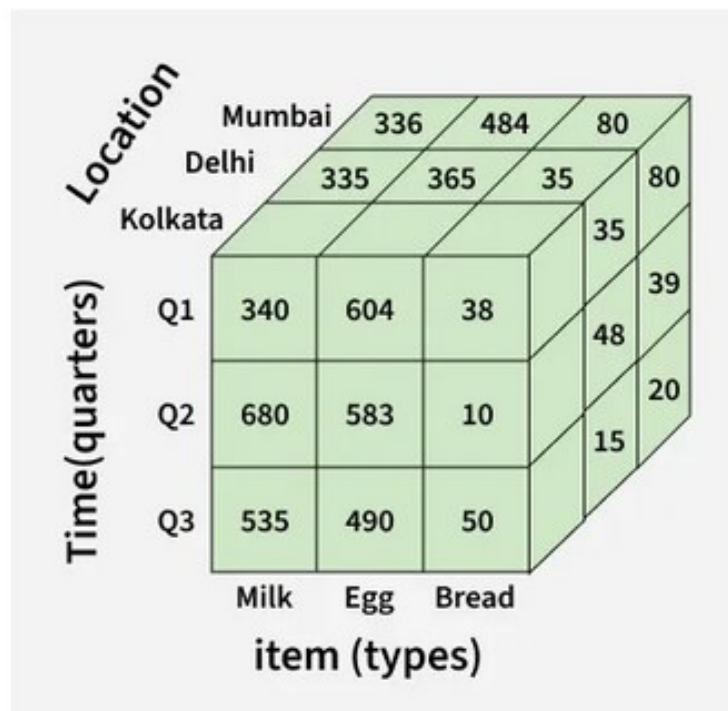
In the above given presentation, the factory's sales for Bangalore are, for the time dimension, which is organized into quarters and the dimension of items, which is sorted according to the kind of item which is sold. The facts here are represented in rupees (in thousands).

Suppose we need to add multiple cities (for e.g. Kolkata, Delhi, Mumbai) then the table can be extended as:

Time	Location="Kolkata"			Location="Delhi"			Location="Mumbai"		
	item			item			item		
	Milk	Egg	Bread	Milk	Egg	Bread	Milk	Egg	Bread
Q1	340	604	38	335	365	35	336	484	80
Q2	680	583	10	684	490	48	595	594	39
Q3	535	490	50	389	385	15	366	385	20

2D representation of data

The same data above, can also be represented in the form of three dimensions as given below:



3D representation of the same 2D data